

1. Complete before downloading

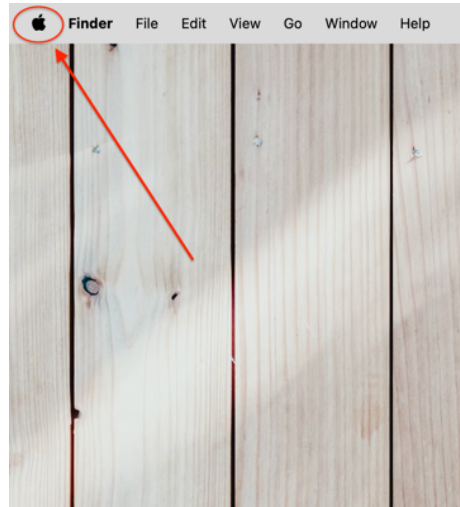


fig. 1

1. To check that your Mac meets the minimum requirements, first click on the Apple logo in the top left corner of the screen (*fig. 1*) and click “About this Mac”
2. From there double check that you are running macOS Ventura (macOS version 13.5 or later). The current latest version of macOS is macOS Sonoma (v14).
3. To check storage on the system click on “System Settings” in the dropdown menu that opens from clicking on the Apple logo in the top left corner, click on “General” and then scroll down to the very bottom and click on “Storage Settings” and make sure you have around 50 GB free to accommodate for installing the software and installing a sound library that is suitable for you.

You need 6 GB for the app itself, and if you’re a musician looking to make music, you will want the extra 44 GB free, to download VST synths, guitar effects, and other important plugins (such as compression and EQ).

4. Under the “General” tab (on the right) of “System Settings” there should be a description of your memory capacity. Depending on your model of Macbook Pro, Mac Desktop, or Macbook Air, you will want about 4GB of RAM (8 GB+ recommended) and this description of memory is your RAM size.
5. Make sure your processor is an M1, M2, or M3. This description will be under the “General” tab as well.

2. Download / launch Logic Pro X



fig. 2

1. Open the **App Store** on your Mac. (icon *fig. 2*)
2. Search for Logic Pro X.
3. Click Buy and enter your Apple ID and password to download the software.
4. Wait for the download and installation to finish (*fig. 3*)



fig. 3

5. Open up Logic Pro X in **Applications** on your Mac

6. When prompted to download a sound library, check mark “Essential Sounds” and then click “Install” (*fig. 4*)

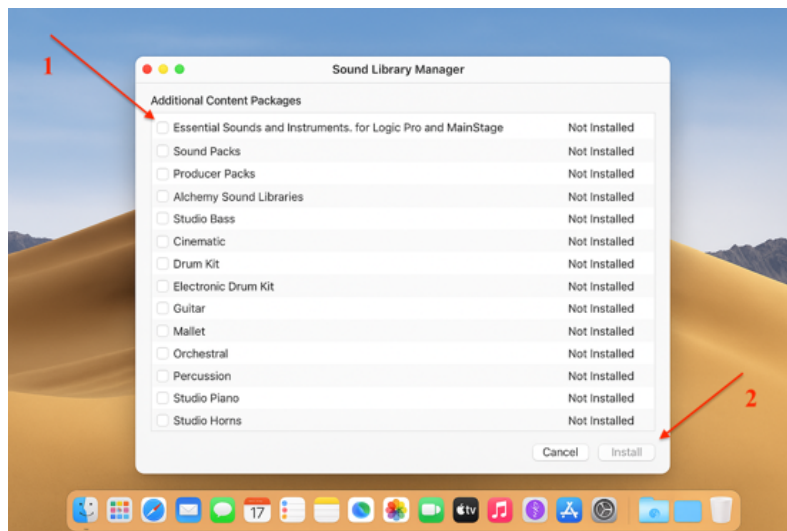


fig. 4

3. Open and save a file in Logic Pro X

1. Once the software is open, double click on “Empty Project” (fig. 5)

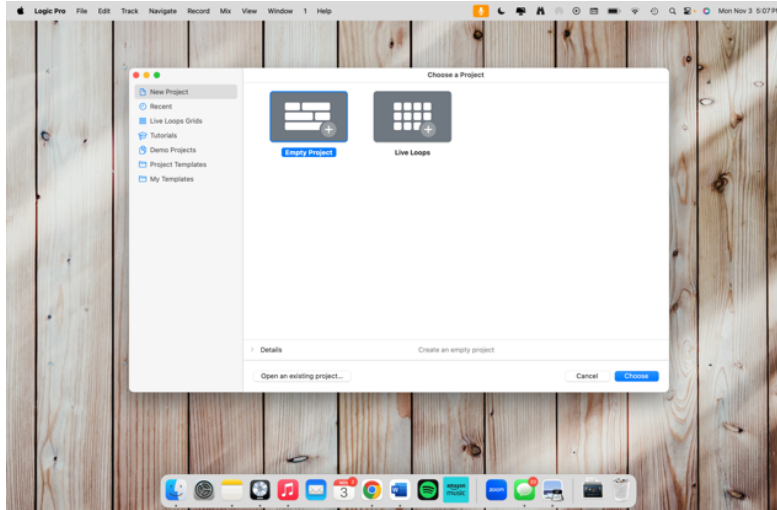


fig. 5

2. Then click on “Create” (default settings are fine) (fig. 6)

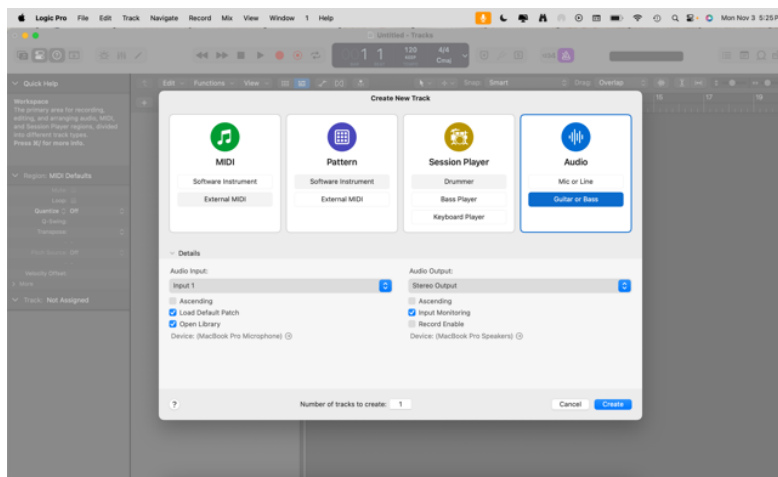
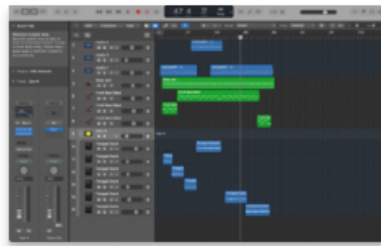


fig. 6

3. Click “File” on the top ribbon then click “Save” to save the project somewhere on your laptop’s hard drive. Notice the icon (fig. 7) that the Logic project has where you saved it (you can navigate to it through **Finder** to look at it). For example purposes, we named this file “Song 1”



Song 1

fig. 7

4. Save tempo information in the file of your to-be creation

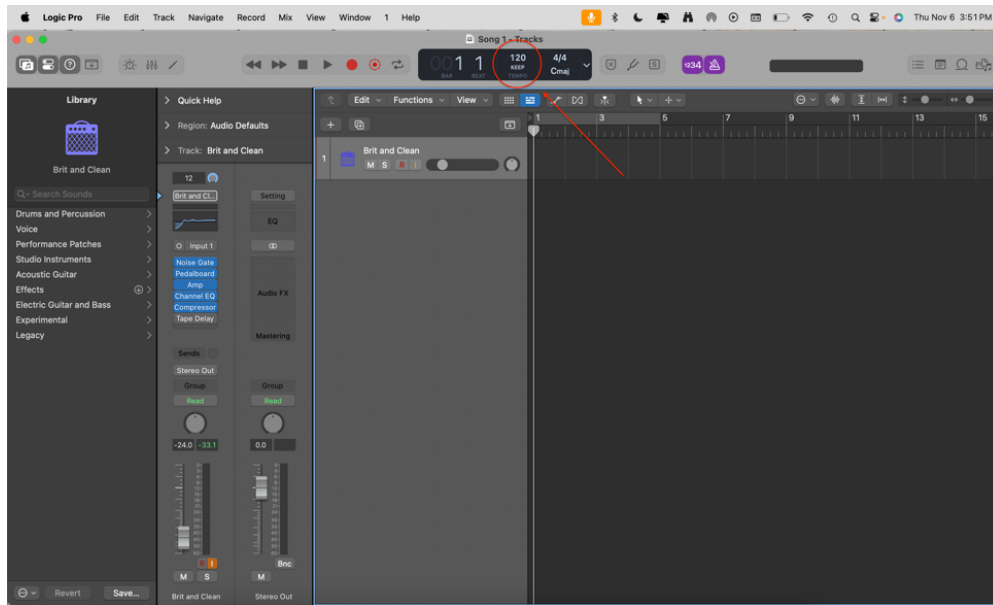


fig. 8

1. Click on the tempo (120) part of the Control Bar (fig. 8) to adjust your tempo to a suitable tempo for the song you're going to create. [Musicca Metronome](#) is a great place to tap with a keyboard key to the tempo of your song to ascertain its BPM (beats per minute), from there, once the file is synchronized to an appropriate BPM, the measure grid will adjust to create four beats for every measure in accordance with the BPM.
2. When you double click on the 120 in the Control Bar, you can type in the tempo you have for your song and go from there. Alternatively, you can move your cursor up or down once you have selected the tempo information to hone in on your intended tempo for the song manually.

You can skip this step if you're recording audio for a film or podcast that isn't related to music-making.

5. Record

1. Click “Track” on the top ribbon of the software and click “New Audio Track”. Make sure your input is set to your laptop mic and you have some headphones plugged in. Due to inevitable audio lagging within the software, you cannot use bluetooth headphones for this part of the user manual. You’ll need some corded headphones for your recording project purposes.



fig. 9

2. To set your input to the laptop mic, click on “Logic Pro” in the top ribbon then click on “Settings” then click on “Audio” (fig. 9.1).

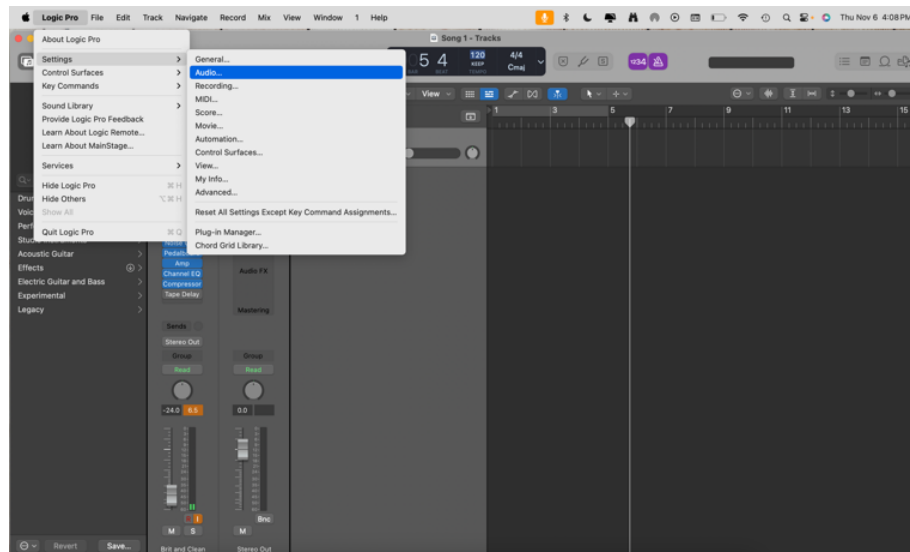


fig. 9.1

- Click on “Output Device” and choose “External Headphones” from the drop-down menu. Then click on “Input Device” and select “MacBook Pro Microphone” or something along those lines (“Built in Microphone”) (fig. 9.2)

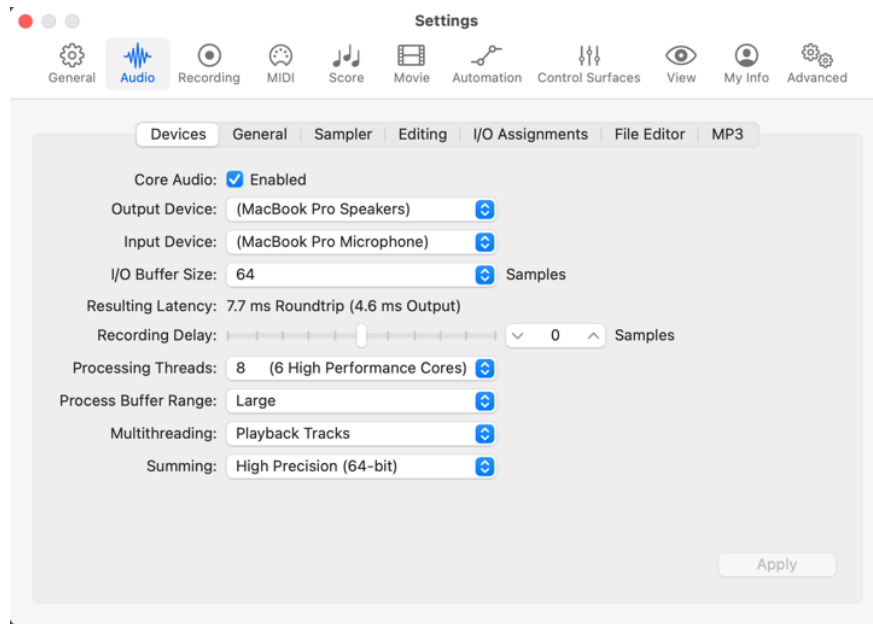


fig. 9.2

- Once you have selected the appropriate settings, you can click “Apply”. From there look at your Logic project again and then with your headphones plugged in, click the **R** button

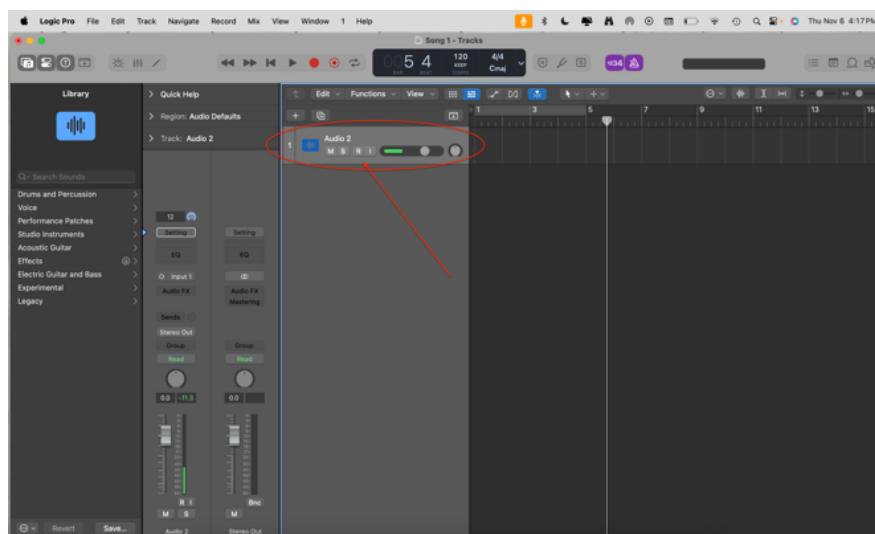


fig. 9.2

Once it starts flashing (alternating with red and gray background blinks), you are ready to record.

5. Now, click the record button at the top of the software's interface on the control bar, and sing something or say something for purposes of testing the software out. I would recommend singing something or humming a gentle tune.

To add a layer of vocals (either singing the same thing to beef up the sound of the entire song's recording, or humming a harmony or back up part that sings a "round"), you can repeat the instructions found in the first paragraph of part 5 of the manual. You are now successfully multi-tracking vocals within the software.

6. Once you have two tracks, you can solo out one of them and play them back (with the "Play" button) by clicking on the white and gray "S" button near the **R** record button.

You can also prepare to mute tracks (to listen back to all your other tracks except for the muted one) by clicking on the white and gray "M" button on the track header.

7. While recording you can right click on the metronome icon (fig. 9.3) on the control bar and then click "Metronome Settings" to configure your ability to hear the metronome while recording on the count-in, while recording, or while playing the music back.



fig. 9.3

Double click on the track header to change the name of your track. For purposes of the manual, "Audio Track 2" will now be named "Vocal Take 1".

6. Trim audio

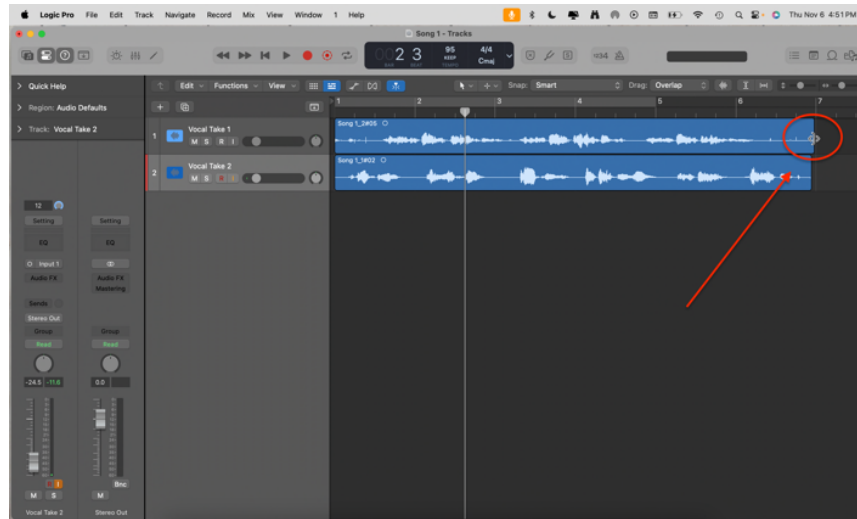


fig. 10

1. Go to the end of one of your audio file recordings, and wait until the mouse toggles to turn into this symbol, and then slide your mouse to the left after clicking, to make the audio file in the “Vocal Take 1” track to be shorter. You can use this technique to trim audio from instrumental or vocal takes that have an unneeded trail at the end. The symbol looks like a staple with a “less than” and “greater than” sign on both sides of it (fig. 10). Once the symbol is toggled, you can click and slide the mouse to the right to shorten your audio file.
2. Move the playhead to a designated area where you want to remove a note you sung or (down the line) edit an instrumental part, to crop out a mistake or simply insert silence for where that track exists as audio information in the Logic project.

For purposes of the tutorial / manual I will move the playhead to measure 3 and delete all audio information until measure 4.

3. Select the audio file until it has a light blue banner on top of it and click ⌘ T (Command + T) then slide the audio file to the right until you reach measure 4. The project should now look like fig. 10.1

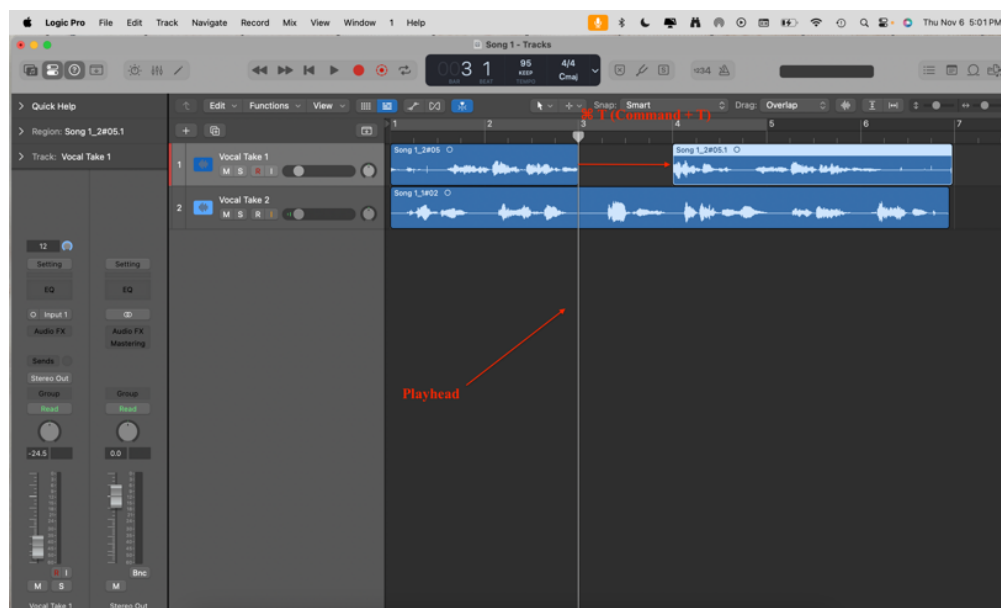


fig. 10.1

This gap you've created in the audio file will now allow play back to skip from playing the harmony of 2 vocal parts to a one part vocal and then back to two part harmony.

7. Create track stacks

This will come in handy down the road when you have multiple guitar parts, synth lines, or vocal takes.

1. To bus together the two vocal tracks that we have created thus far, select the track head for the first “Vocal Take 1” then click shift and then the down arrow on your keyboard.
2. Right click the selection and then click “Create Track Stack” (fig. 11) then click on the bubble next to “Summing stack” then click “Create”

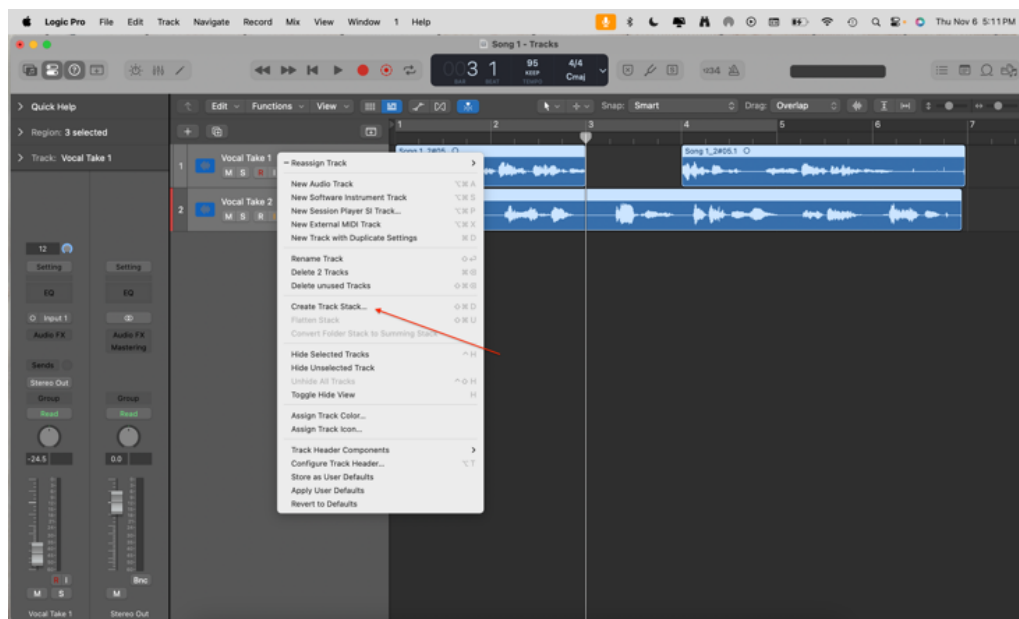


fig. 11

Your project should now look like this (fig 11.1). You can control the volume of the entire track stack from one fader and also add effects to the entire track stack from one place (which we will learn more about later).

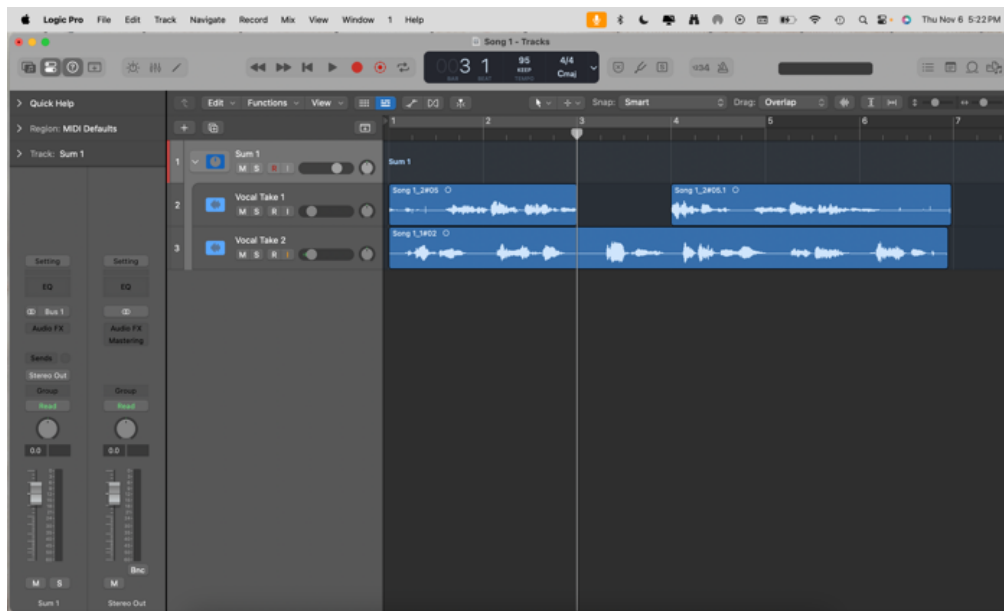


fig. 11.1

3. We're going to change the track stack name to "All the Vocals"

8. Record a guitar or direct-in a piano / synth keyboard

To record a guitar you can use your laptop mic or a microphone that plugs into your laptop (plug and record). Sometimes you need a driver to make the microphone you have work.

1. Let's assume you're still using the laptop mic. Record an acoustic guitar strumming some chords to a new track. You can also use the laptop mic to record an amplified electric guitar.

If you're interested in capturing richer sonics, a good starting microphone (plug and record) to use is the **Blue Yeti** microphone.



fig. 12

2. To direct-in an electric or acoustic-electric guitar, plug the guitar into an interface via a 1/4 inch to 1/4 inch aux cord.

A good starting interface to use would be the **Focusrite Scarlett 2i2 3rd Gen**



fig. 12.1

You can also use the **Focusrite** interface to direct in a piano or keyboard synth to start playing sounds / songs / parts you have created before and record them into Logic and multi-track musical ideas into a masterful song creation.

More expensive mics require an interface like the **Focusrite** to work, so depending on your budget, you can also consider buying a different mic for higher quality audio when recording vocals or acoustic instruments (that aren't able to be plugged in directly into the interface). For example, a violin or trumpet, would ideally be recorded with a microphone like the **Yeti** that can be plugged into your interface.

3. You may need to install a driver to make this interface work with Logic. A manual should come with the interface, teaching you how to download the driver to make it work online.
4. Revisit section 5 of the user manual to configure input settings to accommodate for use of the interface.
5. Start recording guitar tracks.

9. Toggle effects on a guitar or vocal

It might be useful to study these two diagrams before we proceed.

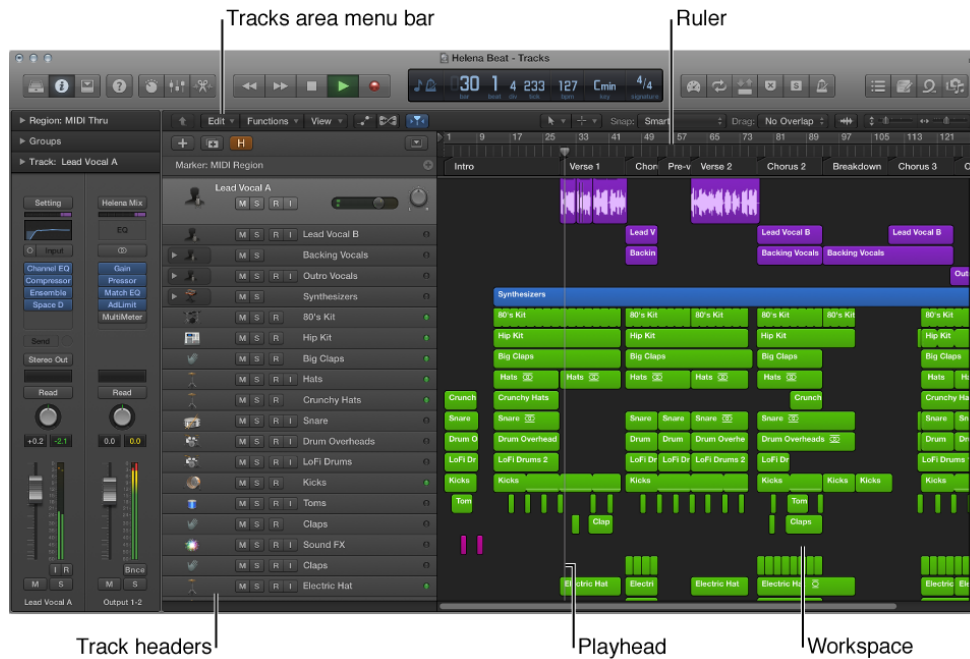


fig. 13



fig. 13.1

A. Chroma Verb

1. Go to the inspector and click on the information for one of your tracks to start toggling effects. Once you click on the effects rig area and choose an effect, I recommend starting with Chroma Verb (fig 13.2).

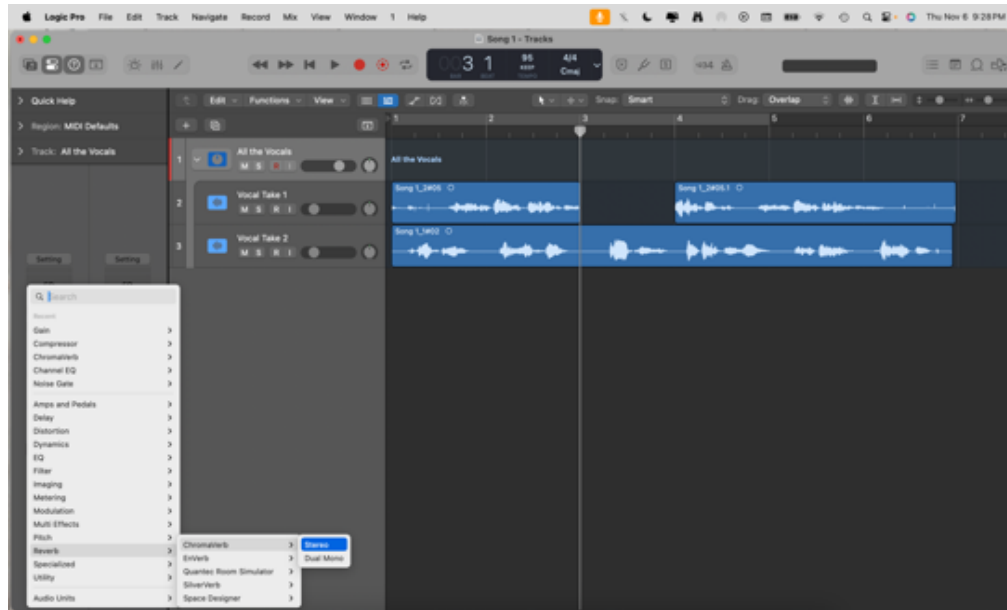


fig 13.2

2. Once you have Chroma Verb designated as the effect you are going to toggle to mysterious effects today (on just another All Hallows's Eve) (fig 13.3). Today we embark on a scare-fun special known to most as Dracula's hot 1800's chamber hall, ready to devour a listener with his gushing bloody teeth. There are so many in-built features to Chroma Verb, that you will likely be engrossed for hours by the space-sounds you can create through customization.



fig. 13.3

3. Right now we are creating a reverb space for both our vocals to the effect of an arena-rock feel, with an intense magnitude of drama and pizzazz. We're hoping for the vocals to capture color and depth that are unheard of by most all people, other than the few worthy listeners of our new track.

4. Let's start with "Big Synth Hall"

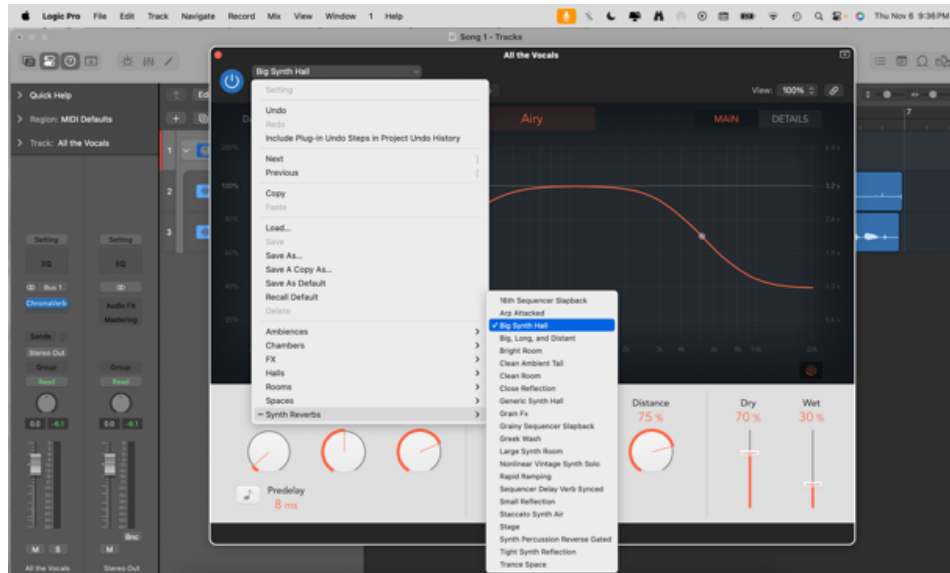


fig. 13.4

5. Toggle everything including EQ to look just like this, and you'll be on your way to creating fun atmospheric reverb.



fig. 13.5

This should sound good for the vocals to the effect of making your beginner vocal sound like it's ready for the big time. A rock arena sound just for you and your voice.

6. 31s of attack should control the creep on the reverb, or the acceleration of the effect as it begins to clip the beginning of an audio sound. Phonemes are captured with intense

precision by the VST effect working its magic through the audio you send it. The engine on this thing is highly reactive and dialing in 31s of the sound existing before the effect begins to attack (oscillation in intensity over the duration of 1.5 seconds). The creep or crawl behavior of cars is reminiscent of what is going on here, when it comes to controlling the part of the sound file that the effect chooses to process and when.

7. The size of the arena is set to 70% to accommodate for loud listeners, easy-going concert goers, and people who have a big sound (imaginatively) existing in their head spatially for what is going on around them. Think walls of speakers but also size that is unbeatable in its ability to deliver a quality transmission to your ears. There's a literal transfiguration of what's happening to the bigness of the room or the size of the outdoor arena in which the reverb exists as it clings to all the objects in the room.
8. The density of the sound can be described only accurately when you play the audio in the workspace with the play button (which you should be familiar by now with, courtesy of the DVD players of the early 2000s). Little balls bounce around at varying speeds in different colors, to represent the sound wave frequencies existing as almost a frequency of atoms filling up different volumes in space. As less molecules exist, the atoms begin to bounce slower, and we realize that a more hyperactive hot pan is better. So we set density to something palatable to our ear. 75% never hurt anyone.
9. 1.5s of decay, outlines the amount of fade the vocal has after it has been processed by the reverb engine. After a note is sung, we hear its echo for approximately 1.5s before moving to the next note, that another ear or a listener has to process while listening. The trail of the reverb the vocal effect is creating is editable. This isn't your average drop of a chip sound. When the pin drops to the floor in this room, its resonance is perceptible by even the most ungullible of ears.
10. When you set the mic distance away from the sound or the initiation of treble in the room, created by none other than your vocal cords, it is important to consider how close you'd want such a vessel to be to your lips. While you are the actor here, emitting the noise, one must also consider the vicinity that an instrument could share to the mic itself.
11. Set the dry to 100% so that the entire raw signal is there from the instrument, and then put "wet" or the duty of the effect of the aquatic properties that reverb bestows to the sonic world, to 70% so that the wetness of the sound is adequately dialed in.

B. Distortion

After recording a guitar using the same methods using steps 5 and 8 in the user manual, start editing its sound. Guitar tones are very precise and shaped to the degree of advanced touch of hand these days. As they always have been—since the advent of guitar effect pedals in the late 60s.

1. Open the pedalboard on one of your guitar tracks by clicking on the inspector after clicking on the track header for the guitar track you have loaded up (fig 14.1).

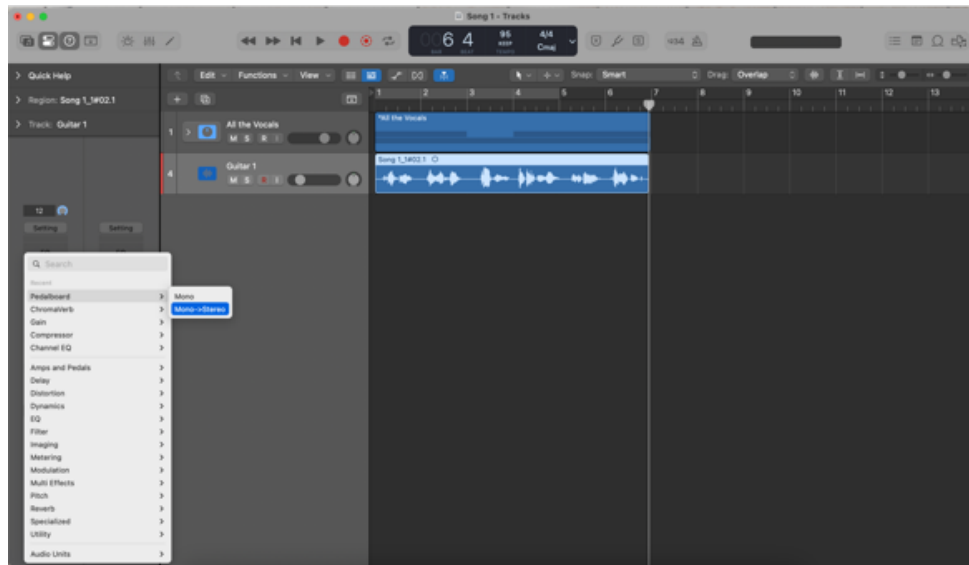


fig. 14.1

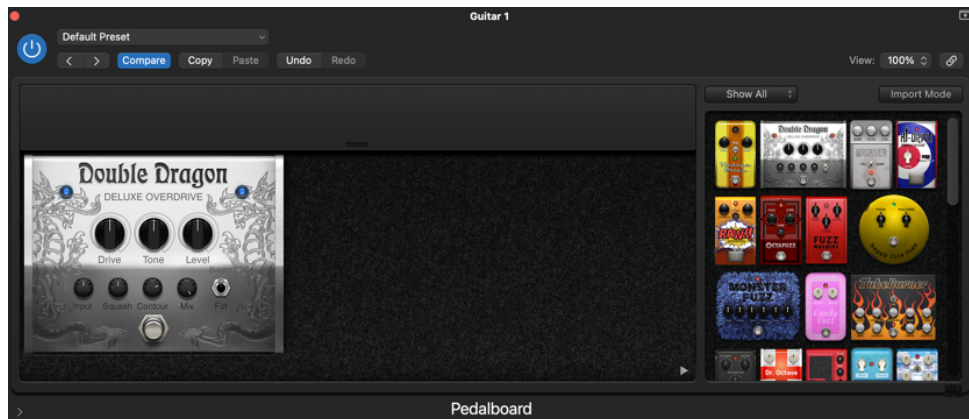


fig. 14.2

2. Drag this pedal onto the board. We will start directing knobs once you've got this step down.

3. For our first guitar we are going to toggle the effects as they are arranged below (fig 14.3). Just move your mouse over the knob you wish to turn and move the cursor up or down.



fig. 14.3

The tone here is thick, aggressive, and clear. Such overdrive would be perfect for surf rock or dream pop purposes. Coupled with some reverb, this could be a powerful tone in the realm of alternative-rock creations.

With these settings:

The **drive** is resultant in a tone with a lot of saturation and crunch. The overdrive here is verging on distortion.

The **tone** is going to be fairly bright with a slight treble boost. Get a loud punchy signal going with this “**level**” knob setting.

Neutral **input** gain lets drive handle the grit.

Light compression applied by the “**squash**” knob allows for some retention of dynamics, so the pick attack is still noticeable.

The **contour** set to half amounts to tight lows and sparkly highs, with less honky mids. The blend with the mix allows for a half clean-tone half overdrive (from the pedal).

In general:

The **drive** knob controls the amount of overdrive increasing saturation and gain progressively as you turn it up.

Tone adjusts the balance frequency of the sound with a higher setting equating to more treble and a lower setting bringing up mids and lows.

The **level** of the effect itself can be used to differentiate between stacked effects. Let's say you bring chorus or tremolo into the mix, you can change this level in order to set the ratio of distortion to other effects that are being thrown onto the guitar's sound.

Higher values on the compression or **squash** knob means you will get a tone that is more compressed and sustain heavy with less attack transient.

A lower **contour** will result in a more "metal" sound and a boosted contour will allow for more "classic rock" sounding overdrive.

The **mix** knob adds distortion to a clean sound allowing you to mix the two (as a parallel processing exercise) to have a tone resultant that has the perfect amount of overdrive added to the initial signal provided.

The **bright** switch will add extra high-end sparkle and bite while an engaged **fat** switch boosts low mids and bass frequencies adding warmth and thickness to the vibration of your strings.

C. Chorus

1. Open up the pedal board from fig 14.1 and then slide the Double Dragon deluxe overdrive off the pedal board work station.
2. Drag “Heavenly Chorus” onto the pedal board work station. You should be able to find this bright blue pedal on the right where the pedals are on display for you to use, drag onto the effects board, and mix with other processing machines.
3. Stomp boxes are unique for adding flavor and rich tonal qualities to pre-existing guitar tracks by processing them as soundwaves that can be mutilated to a wondrous degree of potential for excitement for listeners.



fig 15

4. Dial the Heavenly Chorus pedal in, like so.

With these settings:

The **rate** is turned up to a little more than halfway to allow for a gentle and relaxed warble. With the rate at this level there is no room for seasick bubbling, which will occur if you turn the rate up even more. The end result is wide ambient shimmer, without the intensity of a fast vibrato.

The **depth** is set pretty deep causing a detuned chorus effect, and at this level to the degree of something thick and dimensional.

The **feedback** or the delayed signal fed back into itself adds some resonance and space but avoids metallic or flangey overtones.

The **density** knobs controls how many modulated voices or delay lines are used. The end result is a multi-layered chorus like an ensemble effect rather than a simple 2-voice chorus.

The **bright switch** isn't engaged or flipped off resulting in a warm, darker voicing.

The end result of the tone is something dreamy but grounded like a smooth liquid swirl under your clean tone. The tone is warm and mid-focused and is characteristic of 80s style chorus cleans (U2), dream-pop or lofi.

In general:

The **rate** knob controls the speed of the modulation or how fast pitch and delay lines are oscillating. With a lower rate, there are resultant lush, wide sweeps ideal for ambient chorus. With a higher rate there is fast, wobbling modulation.

The **depth** knob controls the intensity or range of pitch modulation. A low depth gives the tone subtle shimmer and thickens the tone, while a high depth will detune the guitar and cause wide modulation—more dramatic chorus. In simple words this knob will control how deep the sound is wobbling.

The **feedback** feeds part of the modulated (delayed) signal back into the input. A low feedback will result in a smooth sound while high feedback can add resonance and a metallic age approaching flanger (an effect you should try messing around with, but not discussed in this manual). The feedback knob determines how much of the effect re-circulates inside itself.

The **density** knob controls how many modulation voices or delay taps are used or how tightly they are spaced. Low density will result in subtle 2-voice chorus, while thick lush, ensemble-style chorus will result from a high density setting with the illusion of multiple instruments playing together. More density will give you a richer and fuller stereo image.

The **bright** switch adjusts the tonal character giving high-end sparkle and clarity for a modern sound while the bright off is warmer and darker (more vintage, smooth and natural).

The **sync** button when it is on syncs up to the tempo information of the song, creating a modulation rate that is correspondent to the tempo you have selected for the project (see section 4 in the manual). The rate knob is then hooks up to note divisions (1/4, 1/8, 1/16) within the measures of the project.

//

5. Try combining effects and experimenting with the different effects that are available for use within the software.
6. This *Tone Method* book recommendation will help you understand different guitar effects and help you sound like a rock-n-roll king in no time: [The Tone Method](#)

9. Plug in MIDI drums or keyboard

1. Get yourself a MIDI cord that has a MIDI connector on one end and a USB-C male ending on the other (fig. 16).



fig. 16

2. There should be a connecting port on your synth keyboard / electronic piano that will allow you to capture MIDI information from notes you're hitting with Logic Pro X (FIG. 16.1).
3. Connect to a port looking like this one on your electronic drum kit or synth. Connect the other end of the cable to your laptop or desktop Mac device.



fig. 16.1

4. You should get a notification as soon as you plug in the synth or piano keyboard that looks like this (fig. 16.2):

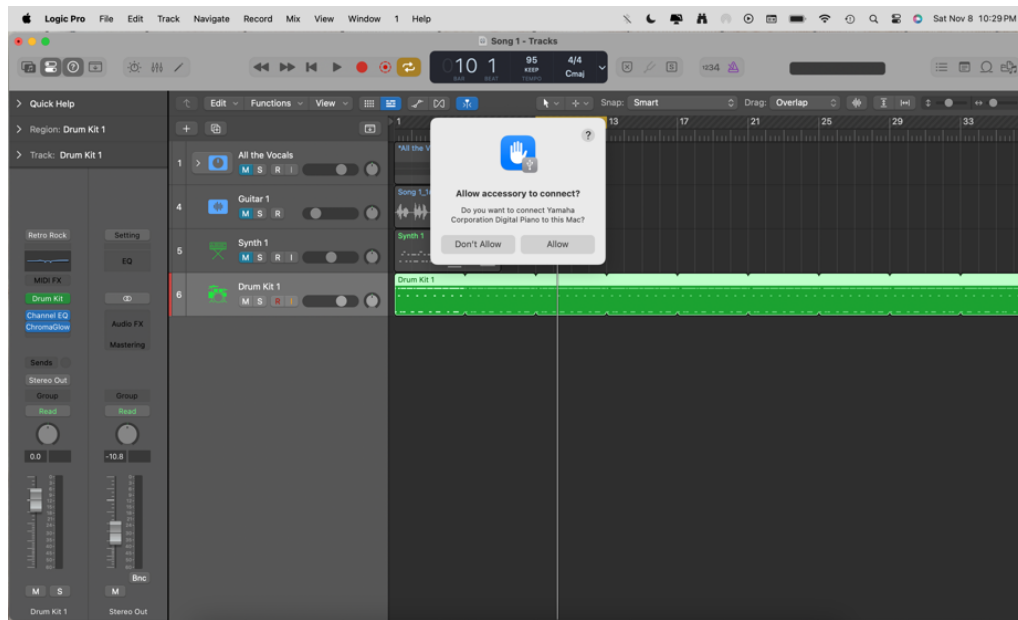


fig. 16.2

10. Use MIDI technology

MIDI technology is unique for allowing you to capture note information / velocity data without dealing with audio information. This allows MIDI captured information to be highly interchangeable, for example you could record some piano chords dialed in to the piano VST (virtual studio technology) sound preset, and after recording shift the whole sound to violins (string section) or fat chip-tune synth chords. A cool feature of MIDI is being able to edit note lengths and positions and note choice as well. Since it's all stored as data, you can edit this information in the MIDI window once the musical composition or part has been recorded. This allows for precise syncopation of EDM chords or synth lines that require perfect timing and cutoff lengths within the song. Like editing a guitar tone, with MIDI, you have a signal to work with to shape synths with powerful onboard and external (third-party) synth modifying software. Creating synth sounds is very sophisticated stuff, and while it's a subtle art, we are only going to dial in one synth sound for the purposes of this manual. Feel free to experiment with presets that come with the software though.

MIDI technology is also great for drums. If you miss a beat, you can slide the snare or kick drum information over by millimeters within the Logic Pro X MIDI interface in the piano roll, which is the window that opens up with MIDI recorded information. You can also fully customize drum sounds with different VST preset sounds. If the drums are recorded to one track you can customize the levels of the individual parts of the drum kit. If you separate every part of the drum kit onto its own track, you can have a fully custom kit. You can EQ drum sounds individually to add character or tone to a snare, kick drum, cymbals, or hi-hat. You can add a little bit of reverb to a snare to make it extra splashy in your mix.

The ES2 (Emagic Synth 2) synth machine is powerful for being able to craft masterful synths that will add sonic depth and texture to your songs. Retrosynth and Alchemy are another two in-built synths / VST software synth shapers that are great for creating the synth of your dreams.

1. Open up a new software instrument and let's get started (fig. 17).

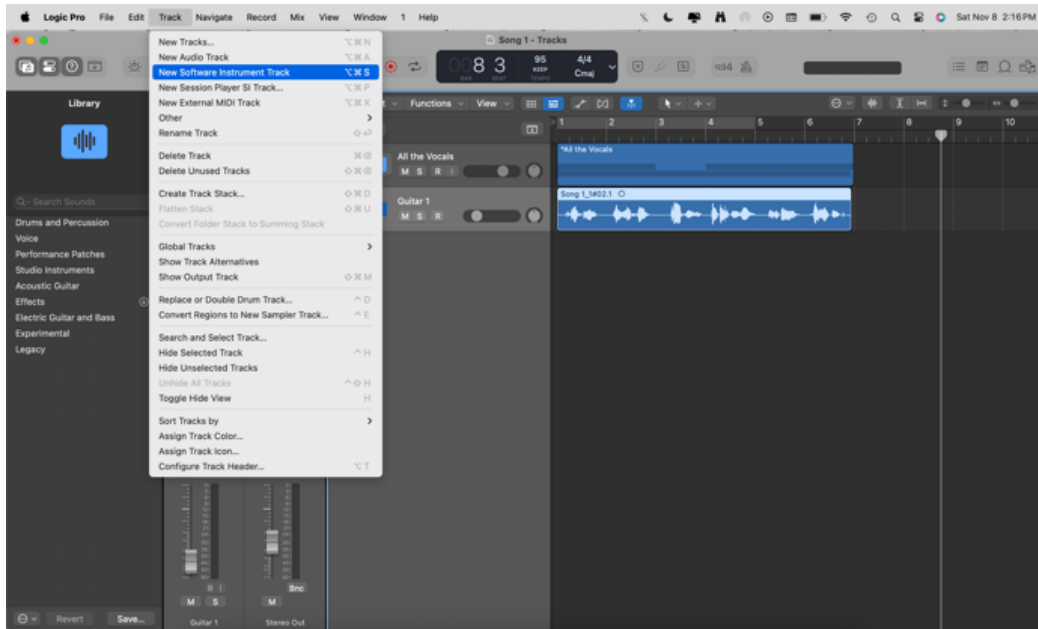


fig. 17

2. We're going to rename "Inst 1", "Synth 1" by double clicking on the track header.
3. Then click on "Synthesizer" in the Library section of the software's interface.

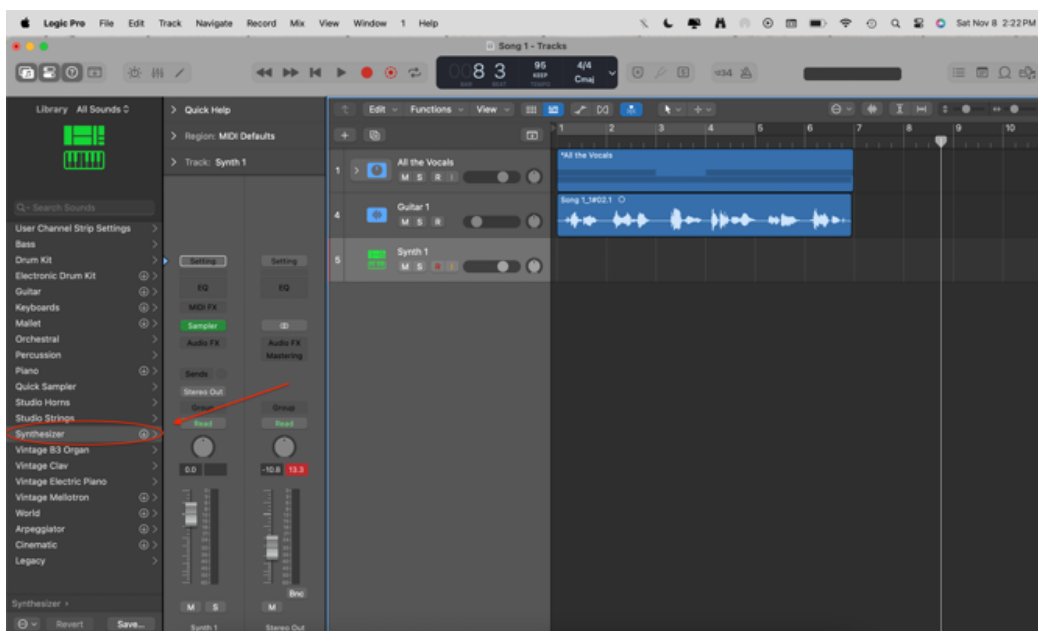


fig. 17.1

4. After you click on “Synthesizer”, click on “Lead” and then “70s Analog Lead”. Play around with this sound with your keyboard plugged in. Make sure your external headphone port is also hooked up to a set of headphones or speakers, so you can hear the synth emit it’s out-of-this-world sound.
5. Record something—maybe a simple lead line and some chords. After the recording step is complete, we can start modifying the tone of the synth.

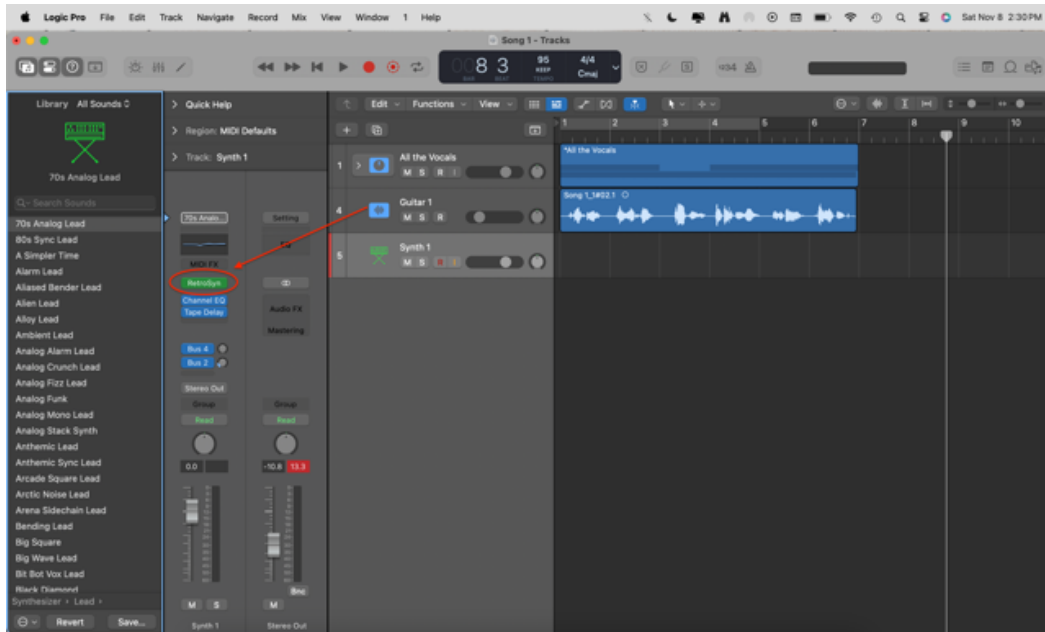


fig. 17.2

6. After completing the last step, click on “RetroSyn” and we can start shaping the synth with the engine that opens up.

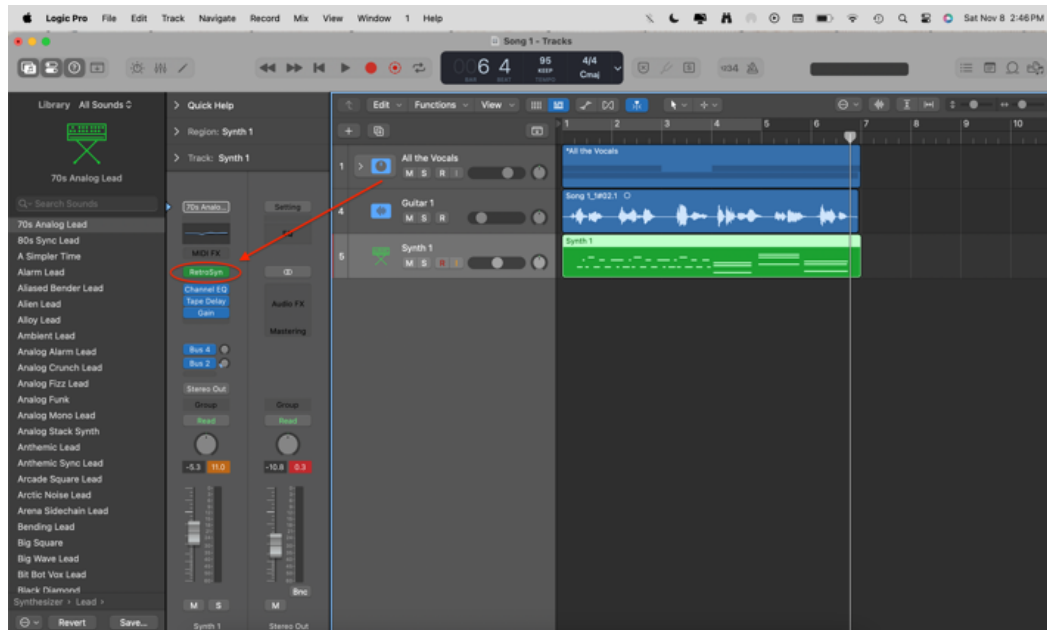


fig. 17.3

7. You're going to modify the synth to look like this (fig. 17.4).

8. To do this you will change “Shape 1” from saw to square. Then turn the “Shape 2” knob until it is halfway between the two square wave shapes. Turn vibrato to a little more than 9 o’ clock. Drag the LP24dB Lush (Fat) wave shape to look like fig 17.4 and then make the sin level a little past 11 o’ clock. You can optionally mess with “Glide” after looping the synth part you have written and recorded. Glide creates a legato blending effect between notes as they transition from one to the next.



fig 17.4

Typically pads can hold chords well and lead synth presets do not play chords, since they can only emit one note at a time.

9. Now let's try dialing in a drum set. Assuming you have an electronic drum kit, this step shouldn't be too hard to follow.

A good kit to start on is the Roland TD-07K VX V-Drums



fig. 17.5

10. Plug the MIDI cord into the drum kit (fig. 16.1) and follow the first step in section 10 (fig. 17) to open a new track.
11. Once you have the new software instrument track open, rename the track “Drum Kit 1” then go to the Library section of Logic’s interface and click on “Drum Kit” (fig. 17.6).

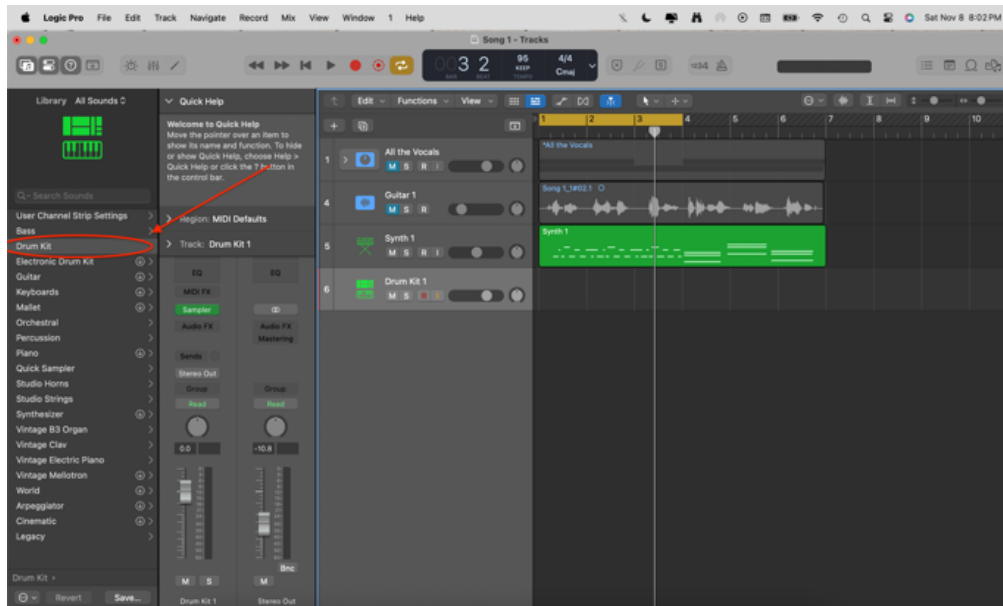


fig. 17.6

12. For purposes of this tutorial, we will select the “Retro Rock” drum kit. This is an acoustic kit that is designed to capture the vintage punch and character of classic rock recordings from the late 60s.
13. If you don’t have a drum kit, but still want to attempt recording a drum part for your song, you can take these steps.
14. Open a musical keyboard with $\text{⌘} + \text{K}$

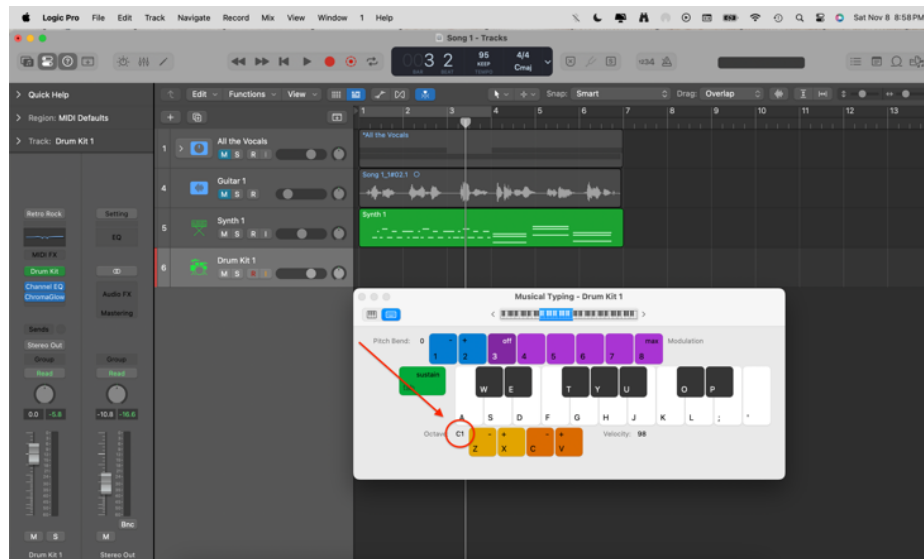


fig. 17.7

15. Make sure the blue at the top of the musical keyboard is selecting octave C1 by clicking in different sections of the long piano at the top.
16. Make a simple kick / snare beat with your keyboard after hitting record by using the “A” and “D” keys on your physical keyboard.
17. To connect your Roland drum kit to the software, first note that after plugging in the drum kit, the software should notify you that you have connected V-Drums by saying “A connected drum controller has been detected.”

18. First click on “Drum Kit” in the inspector area (fig. 17.8).

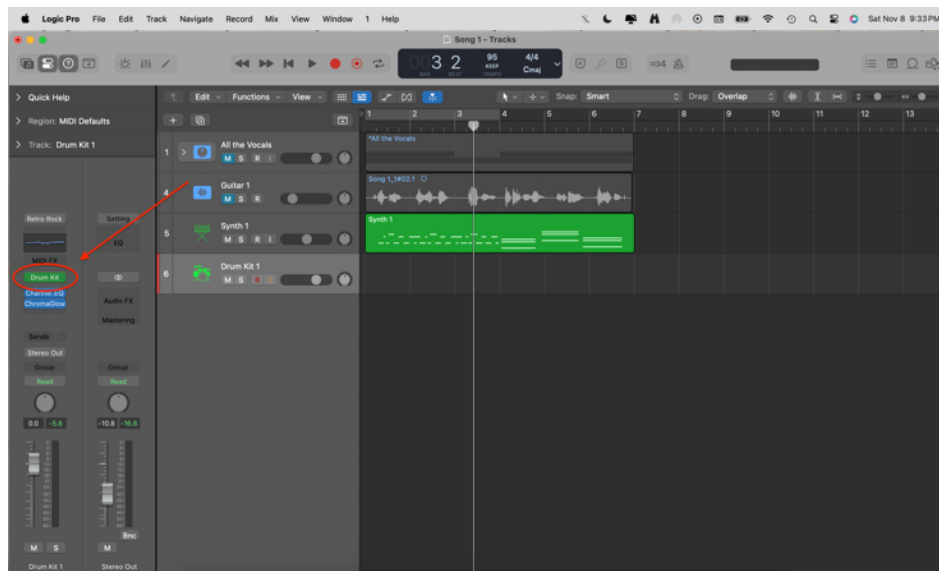


fig. 17.8

19. Once the Drum Kit Designer is open, click on the arrow facing right to drop down settings.



fig. 17.9

20. From the “Input Mapping” selection options, click V-Drum to activate your MIDI drum kit (fig. 17.10).

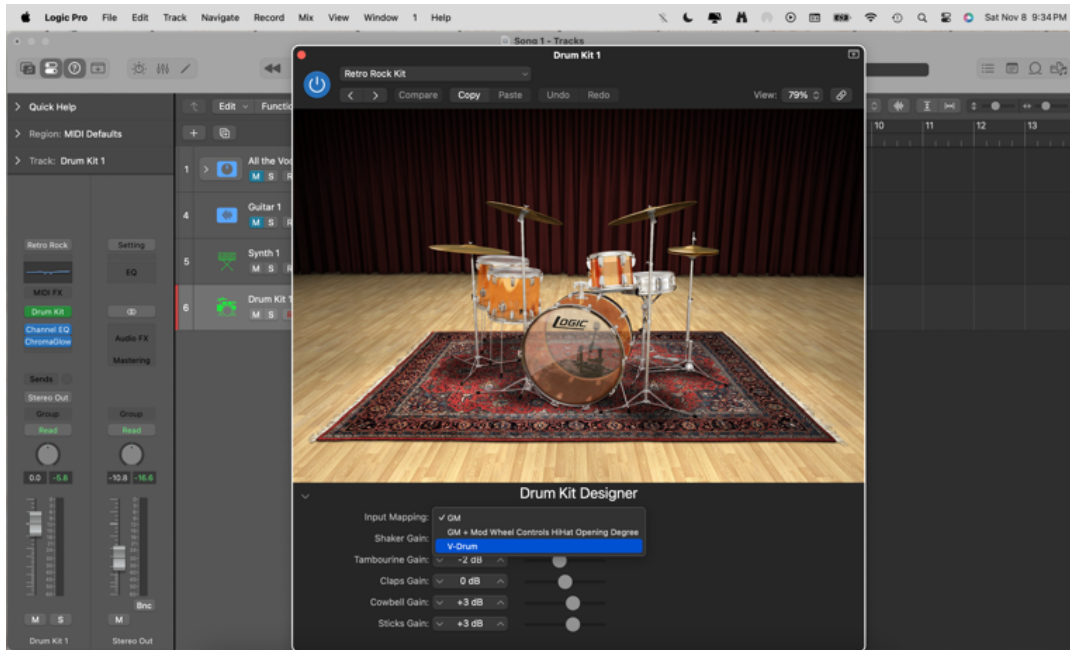


fig. 17.10

21. You should now be able to hit record and with your drum kit plugged in, and be able to track live drums. I would suggest turning the metronome on before you begin.

22. Once you have a drum part you like recorded, trim it to 4 bars and loop it. To do this first move the cursor over the corner of the recorded MIDI track and once this symbol shows up (fig. 17.11), stretch the track to the right.

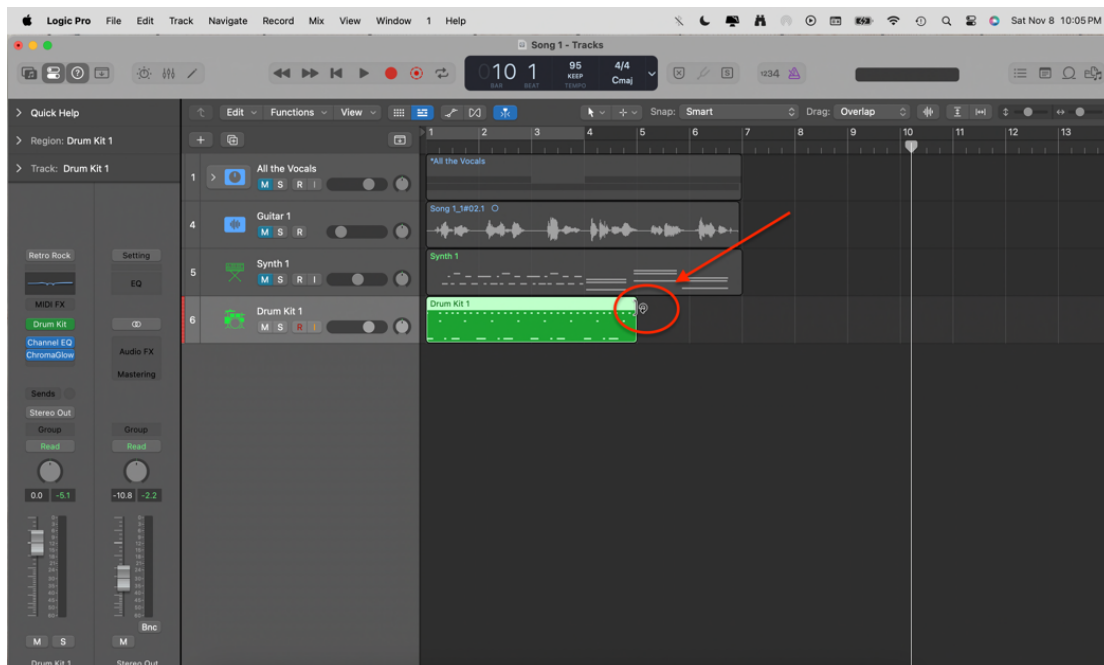


fig. 17.11

If you are deciding between 4 bars in a long recorded drum session, it would benefit to understand the looping tool that is in-built to Logic to find a part of the drum take that is locked in. You can then trim the take to just the four bars that you like and drag it to the beginning of the music session.

Above the ruler you can highlight measures and they will turn yellow (fig. 17.12). Once you press the play button and looping is turned on, it will loop back the highlighted measures. If you click again on the yellow highlighter it will turn dark gray, meaning you have exited loop mode.

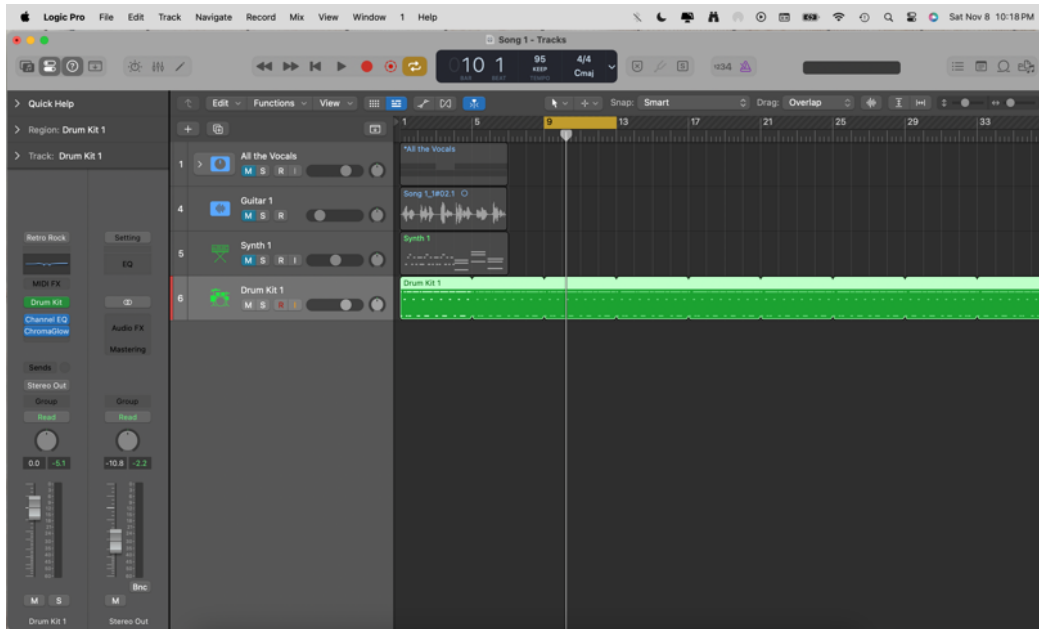


fig. 17.12

23. Once the drums are recorded, you can run levels on the different sounds by clicking on the “Smart Controls” button on the “Control Bar”. Make sure the “Drum Kit 1” track is selected before clicking on the “Smart Controls” button (fig 17.13).

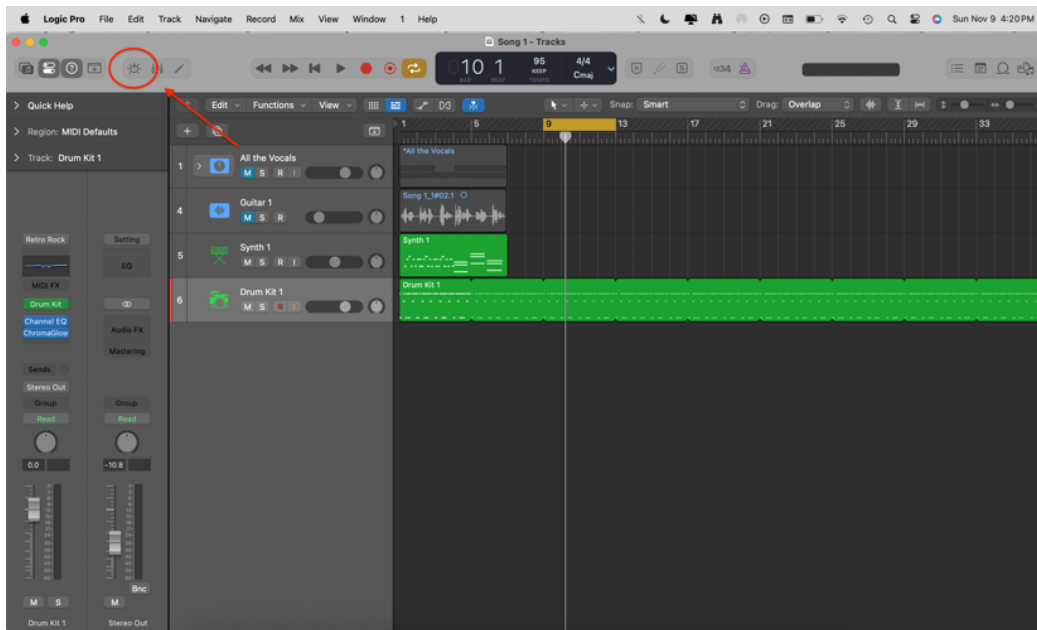


fig. 17.13

After smart controls are open, you can dial in some tasty levels (fig. 17.14). Typically it's customary to have the snare be the same level as the volume of the guitars in the song. The level of the hi-hat can be dialed in while the song is playing to the degree of “racketiness” that you want to make with the metallic hits or precise direction for the syncopation. Use a soft hi-hat for folk songs and a closed hi-hat for more rock-driven songs. The level on the cymbals can be used to enhance the sonic breadth of the song. Choruses typically have cymbals in the background creating atmosphere. You can pan the cymbals a little left, to add depth to the world of the song you're creating.

24. I also suggest experimenting with oscillating the level of the hi-hat in live time throughout the duration of the song to the effect of professional production. Click on the “automation” button to do this (fig. 17.15). You can hit play on the song and slide the fader in time to the levels you wish to hear once you’re in touch mode, make sure to put the track back in “read” mode after you’re done though. This “disappearing hi-hat” will add an element of memorability and catchiness to your song.

It might help to have this control bar key for your software access purposes (fig. 17.16).



fig. 17.14

25. Adjust these smart controls as you see fit with a couple guitars on, maybe a bass guitar and a synth. See what you get. This ability to customize drum levels isn't unique to MIDI drums. Usually a real acoustic drum kit is miced with 6 mics with each part of the drum kit recorded to a different track (with an interface that takes 6 mics), so levels were traditionally able to be changed, even before MIDI drums technology took off.



fig. 17.15

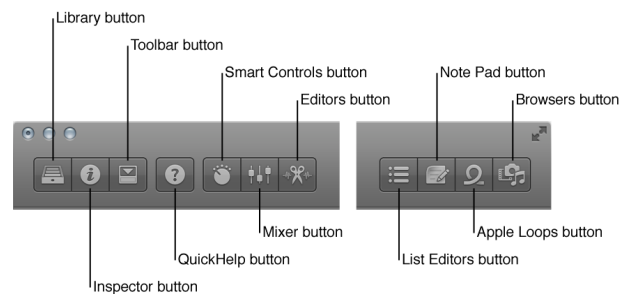


fig. 17.16

11. Mixdown / bounce project to an audio file

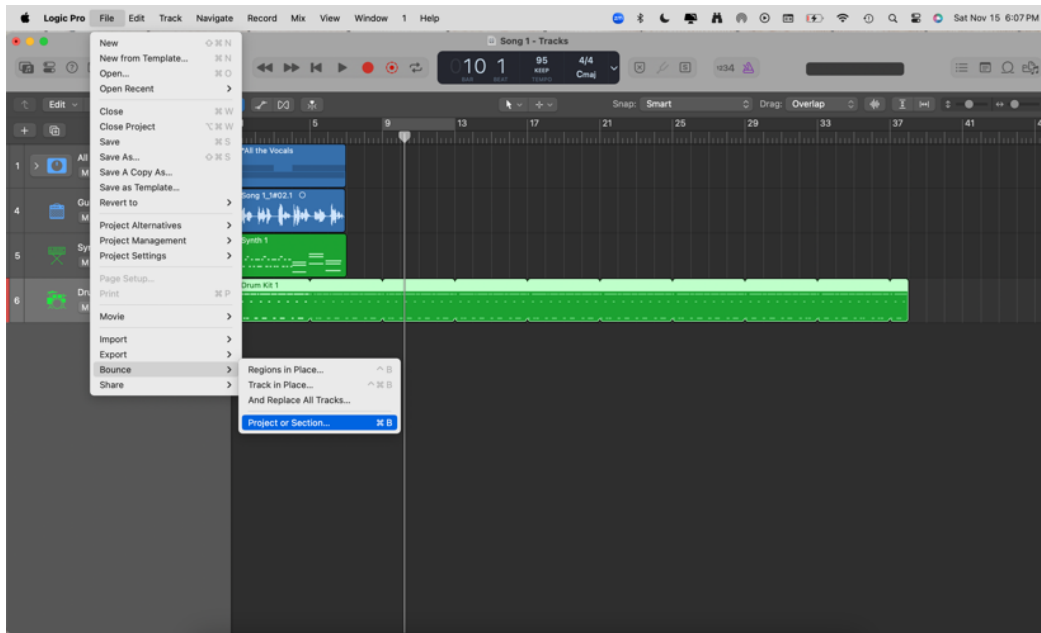


fig. 18

1. When you are done with the project, click on “File” on the top ribbon and then click “Bounce” then “Project or Section”. Alternatively you can hit ⌘+B (Command + B) (fig. 18)

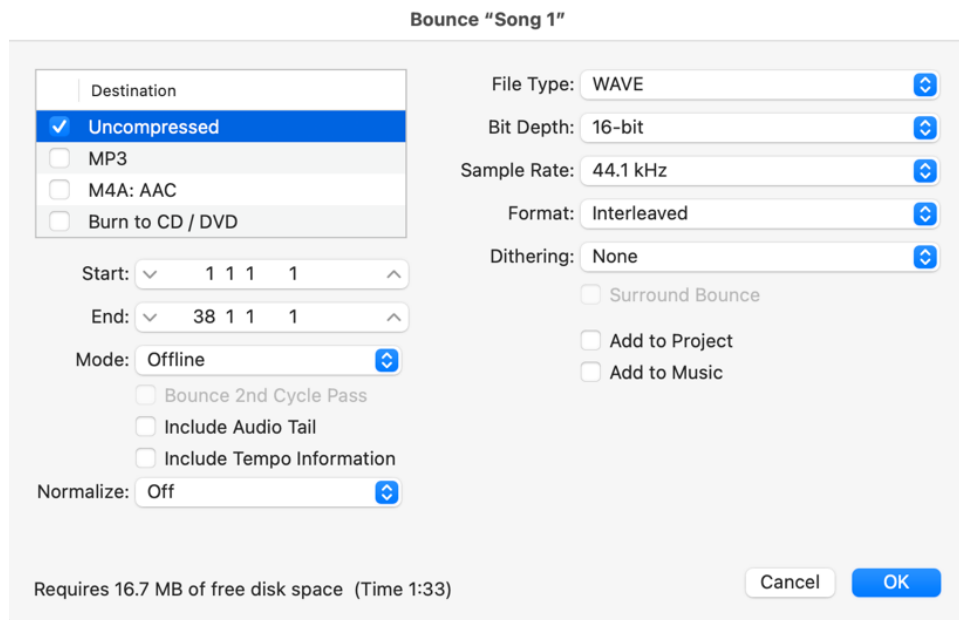


fig 18.1

2. Click the checkbox on “Uncompressed” to yield a high quality WAV file. Enter your start and end measures for the mixdown. The rest of the settings from fig. 18.1 should work well for your audio file creation purposes. After clicking “OK” save the file to somewhere on your hard drive that is convenient.